

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study
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Github Repository	Click Here

1 Objective

- To implement a Decision Tree classifier.
- To extend the Decision Tree into a Random Forest ensemble model.
- To study the impact of hyperparameters on overfitting and generalization.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare single-tree and ensemble-tree models on the basis of Accuracy, Precision, Recall, F1-score and ROC-AUC.

2 Problem Statement

Given 30 numerical features describing the characteristics of cell nuclei present in a digitized image of a fine needle aspirate (FNA) of a breast mass, the task is to classify each sample as **Malignant (M)** or **Benign (B)**. A Decision Tree classifier is trained first as a baseline single-tree model. A Random Forest ensemble is then trained and tuned to examine whether bagging and random feature selection reduce variance and improve generalization compared to a single tree. Both models are tuned using 5-Fold Cross-Validation and evaluated on a held-out test set using standard classification metrics.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC) Dataset
Dataset Source	UCI Machine Learning Repository (https://archive.ics.uci.edu)
Number of Samples	569
Number of Features	30 numerical attributes (10 base features $\times$ 3 statistics: mean, standard error, worst)
Number of Classes	2 (Malignant, Benign)
Missing Values	None (0 missing values across all columns)
Train-Test Split	80% training / 20% testing, stratified on target class (random_state = 42)

4 Exploratory Data Analysis

Class Distribution

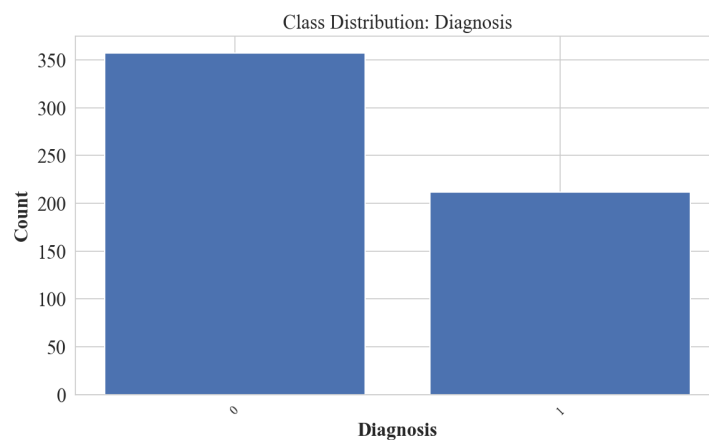


Figure 1: Class distribution of the target variable (Diagnosis)

Inference: The dataset shows a moderate class imbalance, with benign cases outnumbering malignant cases. This is a common pattern in diagnostic datasets, since most screened samples turn out to be non-cancerous. Because the imbalance is not extreme, stratified train-test splitting was used to preserve class proportions across training and test sets. Weighted averaging was used for precision, recall, and F1-score to account for the imbalance during evaluation. The imbalance also motivated the use of ROC-AUC as a complementary metric, since it is less sensitive to class skew than accuracy alone.

Correlation Analysis

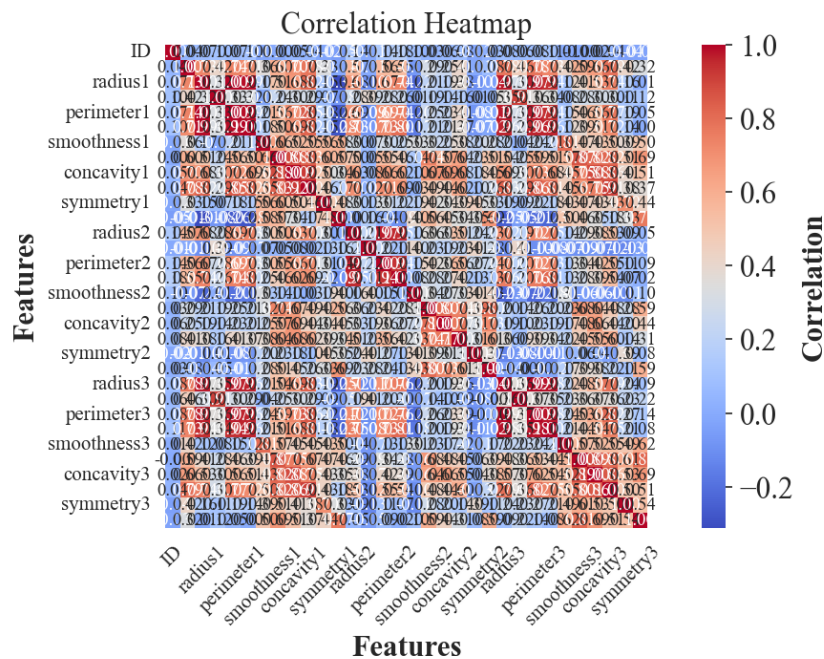


Figure 2: Correlation heatmap of numerical features

Inference: Several feature groups show very high mutual correlation, particularly the radius, perimeter, and area measurements at each of the three scales (mean, SE, worst), which is expected since these are geometrically related quantities. This high multicollinearity suggests that tree-based models are a good fit for this dataset, as they are not sensitive to correlated features in the way linear models are. It also indicates that dimensionality reduction or feature selection could be applied without much loss of information. The correlation structure motivated inspecting individual feature distributions (e.g., perimeter1) before modeling.

Feature Distribution

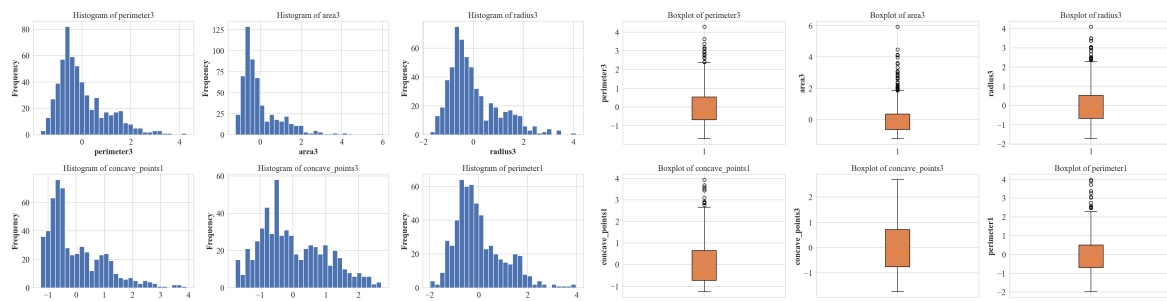


Figure 3: Distribution and spread of the perimeter1 feature

Inference: The histogram of perimeter1 shows a right-skewed, unimodal distribution, with most values concentrated in the lower-to-mid range and a longer tail toward larger perimeters, which typically correspond to malignant samples. The boxplot confirms this

skew and reveals a number of outliers above the upper whisker. Since `perimeter1` was selected as one of the highest-variance/most informative features, its distribution gives insight into how well the two classes might separate along this axis. These outliers do not need aggressive removal since tree-based models are robust to them, but they were retained and instead handled implicitly through the recursive partitioning process of the tree.

5 Data Preprocessing

- **Missing value handling:** No missing values were present in the dataset (confirmed via `checkMissing`); numeric columns would otherwise have been imputed with the median and categorical columns with the mode.
- **Encoding:** The target variable `Diagnosis` (M/B) was label-encoded to $\{0, 1\}$ using `LabelEncoder`.
- **Feature scaling:** All numeric features were standardized using `StandardScaler` (zero mean, unit variance), ensuring no single feature dominates due to scale differences.
- **Train-test split:** An 80–20 stratified split was used with `random_state=42` to preserve the class ratio in both sets.
- **Feature engineering:** No additional feature engineering was performed; all 30 original numerical attributes were retained after dropping the non-informative ID column.

6 Machine Learning Algorithm

Decision Tree Classifier

A Decision Tree recursively partitions the feature space by selecting, at each node, the split that maximizes impurity reduction, measured using the Gini Index or Entropy:

$$\text{Gini}(t) = 1 - \sum_{i=1}^C p_i^2, \quad \text{Entropy}(t) = - \sum_{i=1}^C p_i \log_2 p_i$$

where p_i is the proportion of class i samples at node t . Splits continue until a stopping criterion (max depth, minimum samples per split/leaf) is reached. **Advantages:** Interpretable, requires little preprocessing, handles non-linear relationships and correlated features well. **Limitations:** Prone to overfitting and high variance, small changes in data can produce very different trees.

Random Forest Classifier

Random Forest is a bagging ensemble of N decision trees, each trained on a bootstrap sample of the training data, with each split considering only a random subset of features (`max_features`). Final predictions are obtained via majority voting:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_N(x)\}$$

Advantages: Reduces variance and overfitting compared to a single tree, more robust and generally higher generalization performance. **Limitations:** Less interpretable than a single tree, higher training/prediction cost, and performance gains can plateau or invert if trees are too correlated or too shallow.

7 Experimental Setup

Python Version	3.x
Scikit-Learn Version	1.x
IDE	Jupyter Notebook
Operating System	
Hardware	

8 Hyperparameter Selection

Decision Tree

Search space: `criterion` $\in$ {gini, entropy}, `max_depth` $\in$ {3, 5, 7, 10, None}, `min_samples_split` $\in$ {2, 5, 10}, `min_samples_leaf` $\in$ {1, 2, 4}, evaluated via 5-Fold GridSearchCV.

Parameter	Best Value
Criterion	Entropy
Max Depth	7
Min Samples Split	5
Min Samples Leaf	2
Best CV Accuracy	95.16%

Random Forest

Search space: `n_estimators` $\in$ {50, 100, 200}, `max_depth` $\in$ {5, 10, None}, `max_features` $\in$ {sqrt, log2}, `bootstrap` $\in$ {True, False}, evaluated via 5-Fold GridSearchCV.

Parameter	Best Value
n_estimators	200
Max Depth	10
Max Features	sqrt
Bootstrap	False
Best CV Accuracy	96.70%

Table 1: Decision Tree Hyperparameter Evaluation (Best Combination, 5-Fold CV)

Criterion	Max Depth	Min Samples Split	Min Samples Leaf	Avg CV Accuracy (%)
Entropy	7	5	2	95.16

Table 2: Random Forest Hyperparameter Evaluation (Best Combination, 5-Fold CV)

n_estimators	Max Depth	Max Features	Bootstrap	Avg CV Accuracy (%)
200	10	sqrt	False	96.70

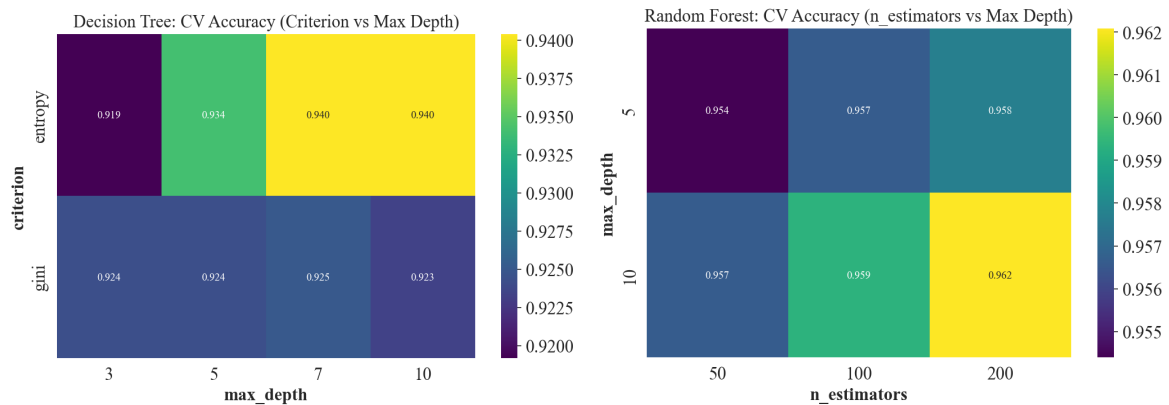


Figure 4: Grid search CV accuracy heatmaps for Decision Tree (left) and Random Forest (right)

Inference: The Decision Tree heatmap shows CV accuracy improving as `max_depth` increases from very shallow trees, but the gains flatten and slightly fluctuate beyond a moderate depth, indicating the onset of overfitting at greater depths. The Entropy criterion performed marginally better than Gini for this dataset at the selected depth. For Random Forest, accuracy improves consistently with more estimators, and this improvement is more stable across depth values than in the single Decision Tree, reflecting the variance-reduction effect of bagging. The best Random Forest configuration used a higher number of trees (200) with unrestricted growth per tree (`bootstrap=False`, `max_depth=10`), showing that ensembling many trees compensates for the risk of overfitting present in a single deep tree.

9 Results

Decision Tree (Tuned) – Test Set

Metric	Value
Accuracy	0.9561
Precision	0.9590
Recall	0.9561
F1-score	0.9555
ROC-AUC	0.9396

Random Forest (Tuned) – Test Set

Metric	Value
Accuracy	0.9737
Precision	0.9747
Recall	0.9737
F1-score	0.9735
ROC-AUC	0.9947

Table 3: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	93.86	90.35	96.49	92.98	95.58	93.85
Random Forest	96.49	93.86	95.61	95.61	96.46	95.61

10 Result Visualization

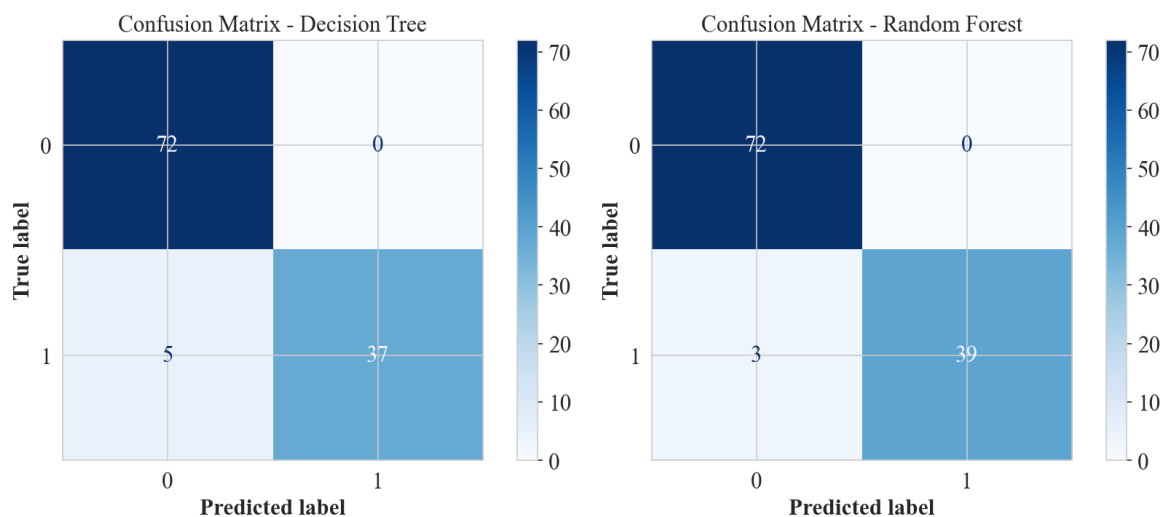


Figure 5: Confusion matrices for the tuned Decision Tree (left) and Random Forest (right) on the test set

Inference: The Decision Tree confusion matrix shows 5 malignant cases misclassified as benign, out of 42 malignant samples in the test set, while all benign cases were correctly classified. The Random Forest reduces this to only 3 misclassified malignant cases, again with zero false positives on the benign class. In a medical diagnosis context, false negatives (malignant cases predicted as benign) are the more costly error type, so the Random Forest’s lower false-negative count is clinically significant. Both models achieve perfect precision on the benign class, but the Random Forest clearly generalizes better on the minority (malignant) class.

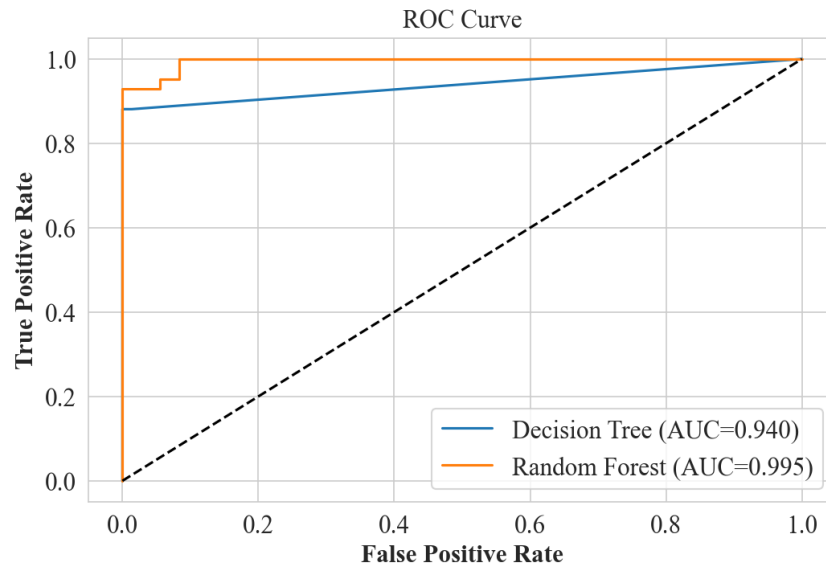


Figure 6: ROC curves for Decision Tree and Random Forest

Inference: The Random Forest's ROC curve lies consistently above and to the left of the Decision Tree's curve, corresponding to its much higher AUC (0.9947 vs 0.9396). This indicates that the Random Forest maintains a better true-positive rate at every false-positive rate threshold, i.e., it separates the two classes more confidently across the full range of decision thresholds. The single Decision Tree's ROC curve is comparatively closer to the diagonal, consistent with its higher variance and coarser probability estimates (since a single tree's leaf-based probabilities are less smooth than an aggregated forest's).

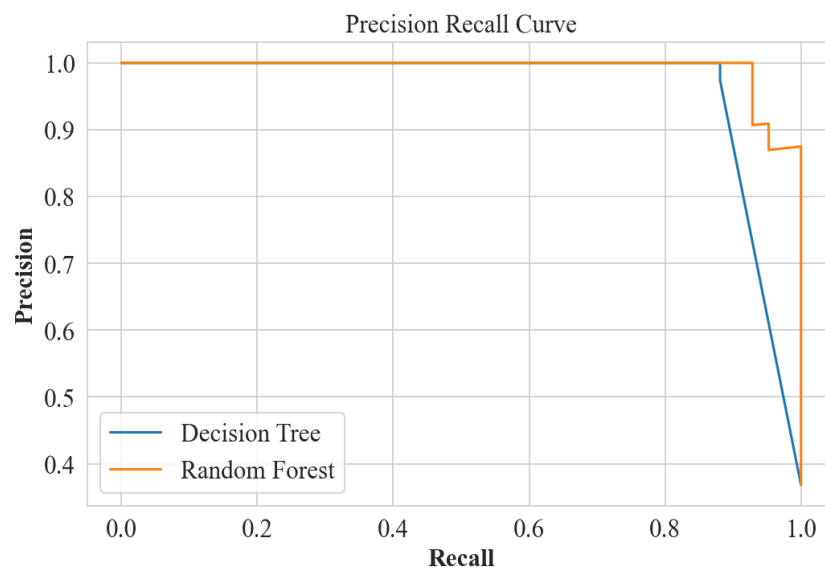


Figure 7: Precision-Recall curves for Decision Tree and Random Forest

Inference: The Random Forest maintains high precision across a wider range of recall values compared to the Decision Tree, whose precision drops off more noticeably as recall increases. This is particularly relevant given the class imbalance in the dataset, since the Precision-Recall curve is more sensitive to minority-class (malignant) performance

than the ROC curve. The Random Forest's curve staying closer to the top-right corner confirms that ensembling improves the model's ability to correctly identify malignant cases without sacrificing precision.

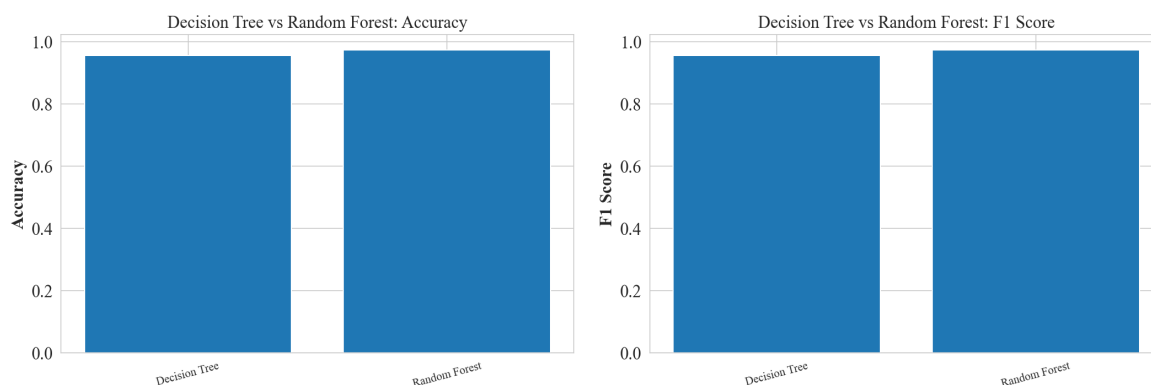


Figure 8: Direct comparison of test-set Accuracy and F1-score between the two models

Inference: Both bar charts show a consistent improvement from Decision Tree to Random Forest, with accuracy increasing from 95.61% to 97.37% and F1-score improving from 0.9555 to 0.9735. This consistent margin across both metrics (rather than a trade-off between them) confirms that the Random Forest's improvement is a genuine gain in generalization rather than an artifact of favoring one class over another. The relatively modest absolute margin also indicates that the single Decision Tree, once properly tuned via cross-validation, was already a fairly strong baseline on this dataset.

11 Discussion

How does tree depth affect overfitting in Decision Trees?

Increasing `max_depth` allows the tree to fit increasingly fine-grained decision boundaries, which improves training accuracy but beyond a certain depth causes the model to memorize noise and outliers in the training data rather than the underlying pattern. This is reflected in the grid search results, where CV accuracy improved up to a depth of around 7 but did not continue improving monotonically at greater depths, indicating a bias-variance trade-off. Restricting depth (along with `min_samples_split` and `min_samples_leaf`) acts as a regularizer that keeps the tree from overfitting.

Which hyperparameter had the greatest impact on performance?

For the Decision Tree, `max_depth` had the most visible impact on CV accuracy, since it directly controls model complexity. For the Random Forest, `n_estimators` had the most consistent positive impact, since increasing the number of trees steadily reduces variance through averaging, whereas `max_features` and `bootstrap` had a comparatively smaller effect once enough trees were used.

How does Random Forest improve generalization?

Random Forest reduces variance by averaging predictions across many decision trees, each trained on a different bootstrap sample and considering a random subset of features at each split. This decorrelates the individual trees so that their errors partially cancel out during majority voting, which is why the Random Forest's 5-fold CV accuracy (95.61% average) was both higher and less variable across folds than the single Decision Tree's (93.85% average, with a low of 90.35% in Fold 2).

Did ensemble learning always improve performance? Why or why not?

In this experiment, the Random Forest outperformed the Decision Tree on every evaluation metric and in 4 of 5 individual CV folds, but the improvement was not dramatic, which reflects the fact that the tuned single Decision Tree was already a reasonably strong model on this relatively clean, well-separated dataset. Ensembling does not *always* guarantee improvement in general; if the base trees are highly correlated (e.g., too shallow, too few features to choose from, or a dataset where a single feature dominates), the variance-reduction benefit of bagging diminishes and the ensemble's gain over a single well-tuned tree can be marginal.

12 Conclusion

Decision Tree and Random Forest classifiers were implemented and evaluated on the Wisconsin Diagnostic Breast Cancer dataset using 5-fold cross-validation for hyperparameter selection. The tuned Decision Tree (criterion=Entropy, max_depth=7) achieved a test accuracy of 95.61% and ROC-AUC of 0.9396, while the tuned Random Forest (200 trees, max_depth=10, max_features=sqrt, bootstrap=False) achieved a test accuracy of 97.37% and ROC-AUC of 0.9947. The Random Forest also showed a higher and more stable average cross-validation accuracy (95.61% vs 93.85%) with fewer false negatives on the malignant class. These results confirm that ensemble learning via bagging reduces variance and improves generalization relative to a single decision tree, particularly in reducing the clinically costly false-negative rate.

13 References

1. Scikit-learn Developers, "Decision Trees," *scikit-learn documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/tree.html>
2. Scikit-learn Developers, "Ensemble methods – Random Forests," *scikit-learn documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/ensemble.html>
3. W. N. Street, W. H. Wolberg, and O. L. Mangasarian, "Nuclear feature extraction for breast tumor diagnosis," *UCI Machine Learning Repository, Wisconsin Diagnostic Breast Cancer Dataset*. [Online]. Available: <https://archive.ics.uci.edu>